SOFC Plant Availability

How to calculate it, and what it is for our March 2026 test campaign.

Contents: 1 The answer · 2 Sorting the stages · 3 The five numbers · 4 Our plant's numbers and open decisions · 5 Where the formulas come from

1 · The answer

Availability is the share of time the plant could have delivered power if someone had asked for it. Add the hours the plant was making power to the hours it was hot and ready to take load, then divide by all the hours in the period. For our March test campaign that gives **about 40 %**.

Why it matters: availability is the number an owner or customer asks first, because it says how often they can count on the plant. It also forces every hour of not being available to get a reason: we chose to stop, or something stopped us.

available 126.4 h = 40 % not available 188.1 h = 60 %

314.5 logged hours, 12-30 March 2026

Every logged hour from 12 to 30 March 2026 (the campaign with all nine trips), sorted by plant stage. Hours with no data are left out.

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Availability = (hours making power + hours hot & ready) ÷ hours in the period

= (4.4 h + 122.0 h) ÷ 314.5 h = 40 %
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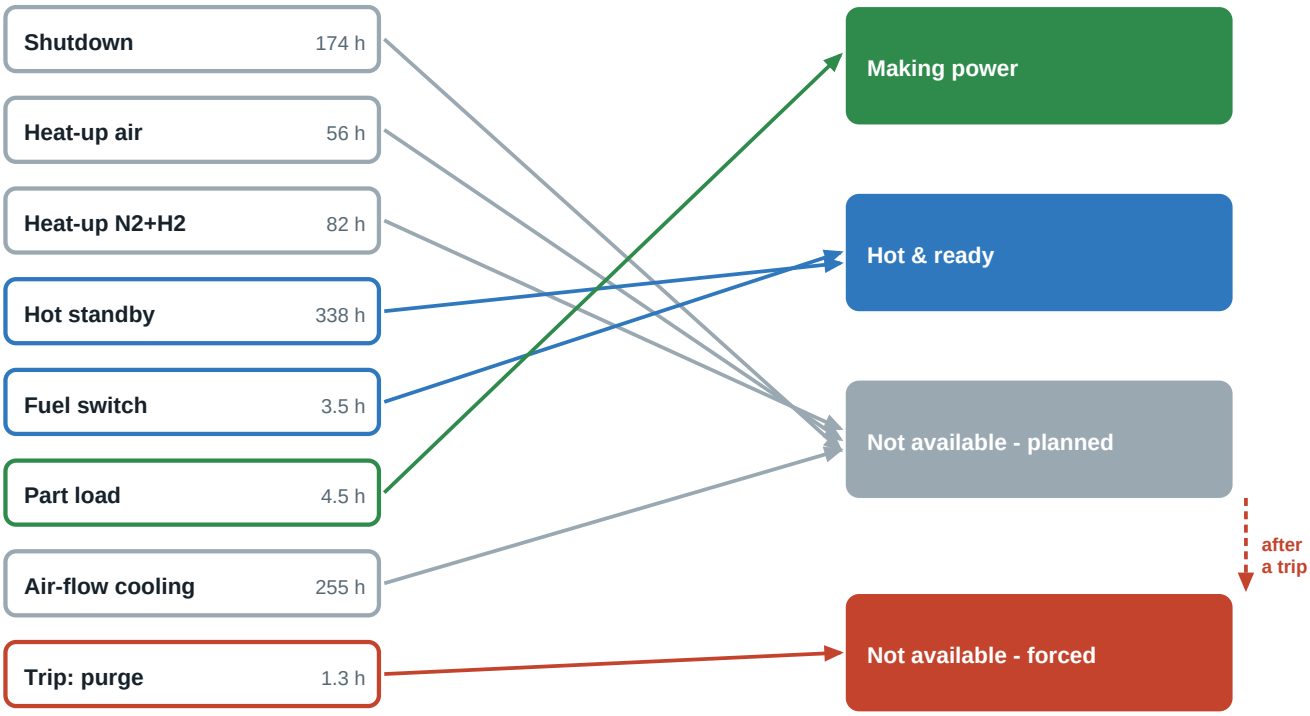
This is the 'availability factor' used for power plants (defined in the IEEE 762 reporting standard; full reference list in section 5).

IF YOU REMEMBER ONE THING

Availability counts the hours the plant *could* deliver power, not only the hours it *did*. Hot standby counts; heat-up, cooling, off and trips do not.

2 · Sorting the stages

The plant log already tells us which of the 8 plant stages the plant was in, minute by minute. Availability needs one more step: decide which bucket each stage belongs to.



Each plant stage feeds one bucket (hours from all 39 logged days). The dashed arrow is the only judgement call: when the plant cools down and restarts because of a trip, those hours can move from 'planned' to 'forced'.

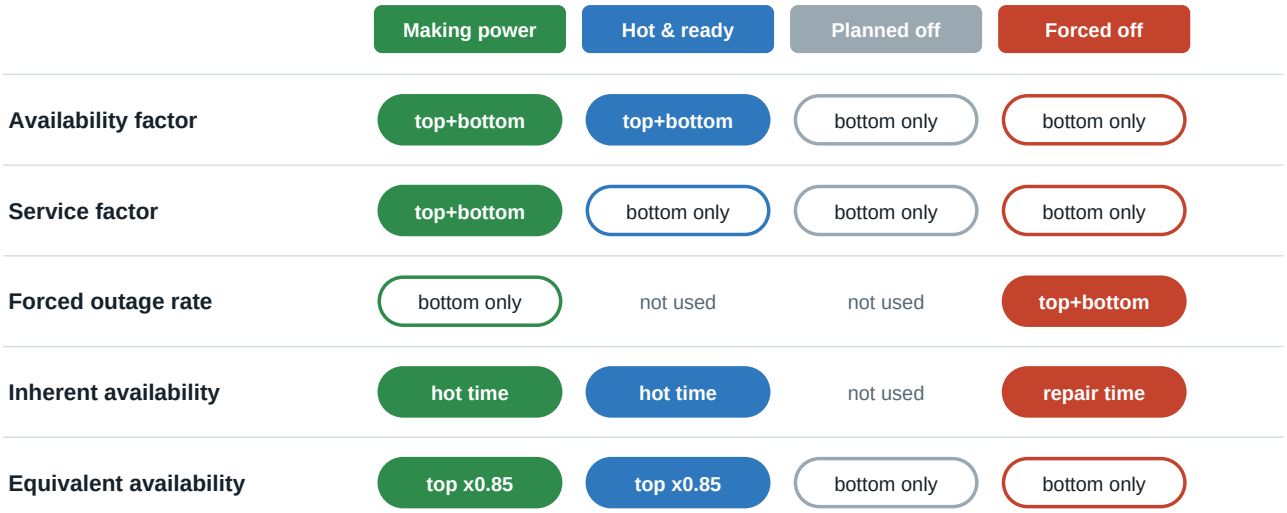
Stage	Bucket	Why
Part load	Making power	Delivers current to the load.
Hot standby	Hot & ready	Stack at working temperature; gas on to power took about 10 minutes (median). Like a car idling at a red light.
Fuel switch	Hot & ready	Changing to natural gas, under 1 A; the first step toward power.
Heat-up air, Heat-up N2+H2	Planned off	From cold the plant needs about 13½ hours to reach hot standby.
Air-flow cooling, Shutdown	Planned off (forced if after a trip)	Stopped by choice, unless a trip caused it.
Trip: purge	Forced off	Always forced.

IF YOU REMEMBER ONE THING

Sort, then add up. Two buckets are available (making power, hot & ready), two are not (planned stops, forced stops after a trip).

3 · The five numbers

Every number is a fraction made from the same four buckets. What changes is which buckets go on top and which go underneath.



Filled pill: that bucket's hours go on top of the fraction (and also underneath). Outline pill: underneath only. x0.85 scales hours by the share of rated power the plant can reach.

Number	Formula	Plain meaning
1 Availability factor	(power + ready) ÷ all hours	Could the plant have delivered if asked?
2 Service factor	power ÷ all hours	How much of the time it actually delivered.
3 Forced outage rate	forced ÷ (forced + power)	When it should run, how much is lost to trips.
4 Inherent availability	MTBF ÷ (MTBF + MTTR)	MTBF (mean time between failures) = hot hours ÷ trips. MTTR (mean time to repair) = average time from trip to hot & ready again.
5 Equivalent availability	availability × (best kW ÷ rated kW)	A plant that can only reach 51 of 60 kW is only 85 % ready. 51 kW is an estimate from part-load data.

What you need from the log: plant stage per minute (have it), trip times (have 9), rated and best reachable power, and a note per stop saying planned or forced.

IF YOU REMEMBER ONE THING

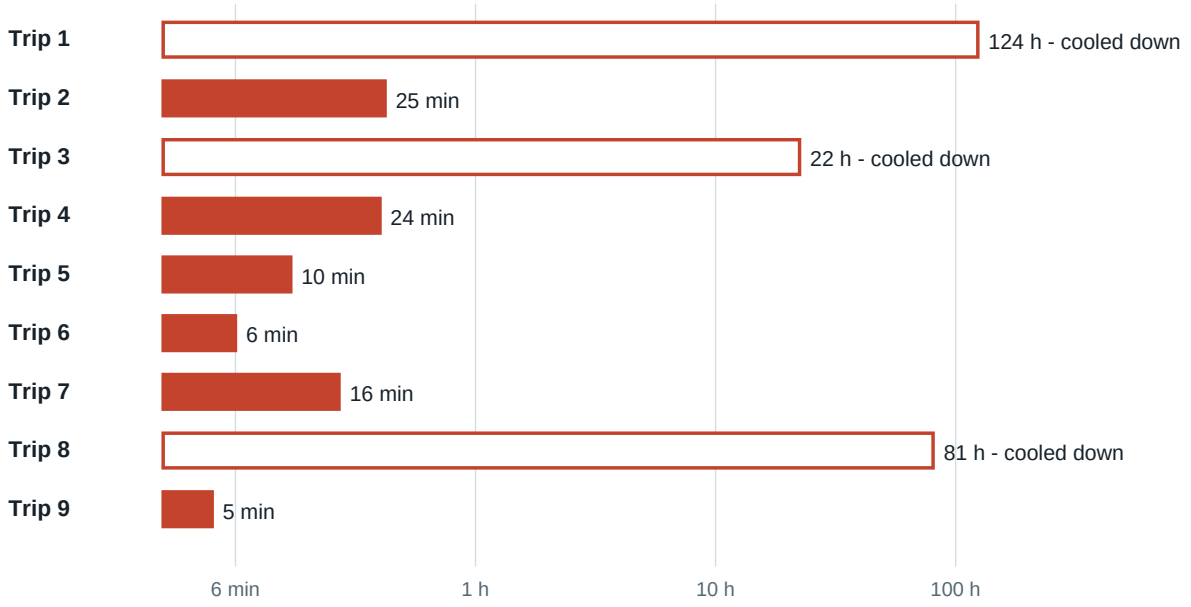
Availability says *how much* of the time the plant could deliver. Forced outage rate and MTBF/MTTR say *why* it was lower.

4 · Our plant's numbers, 12-30 March 2026

Over the 314.5 logged hours the plant was available **40 %** of the time. That number does not change with the trip rule below; what changes is how the other 60 % is blamed.

Number	Strict rule: forced until ready again	Lenient rule: cool-down is a planned stop
Making power / Hot & ready	4.4 h / 122.0 h	4.4 h / 122.0 h
Planned off / Forced off	66.9 h / 121.1 h	185.8 h / 2.2 h
Availability factor	40.2 %	40.2 %
Service factor	1.4 %	1.4 %
Forced outage rate	96.5 %	33.6 %
MTBF (hot hours per trip)	14.0 h	14.0 h
MTTR (trip to ready)	13.5 h	0.2 h
Inherent availability	51.1 %	98.3 %
Equivalent availability (51/60 kW)	34.2 %	34.2 %

The 9 trips: time until hot and ready again



Clock time from each trip to the next minute of hot standby or part load (log scale). Six trips were back in under half an hour. After trips 1, 3 and 8 (outlined) the plant was cooled down and restarted from cold.

The period you divide by also moves the answer

Period	Hours	Availability
12-30 March, logged hours only	314.5	40.2 %
12-30 March, clock hours (5 unlogged days counted as off)	456.0	27.7 %
All 39 logged days (Oct 2025 - Mar 2026)	913.5	37.8 %

This is a test campaign: long cool-downs and off days were part of the test plan, so these figures describe how the plant was used, not what it could achieve in commercial service.

Decisions needed

	Question	Recommendation
A	After a trip, is cooling down and restarting forced or planned?	Strict (forced until ready again), the usual power-plant practice, unless operators note the stop was planned anyway. Moves forced outage rate from 34 % to 97 %.
B	Divide by logged hours or clock hours?	Report both until logging is continuous: 40 % vs 28 %.
C	Does fuel switch count as ready?	Yes; only 3.5 h, barely moves the result.
D	Which power for equivalent availability?	Rated 60 kW; 51 kW best is an estimate. Confirm with a full-load test. With 51 kW: 34 %.

`How to rerun it: data_analysis/scripts/availability.py writes data_analysis/availability/availability_summary.csv and trip_recovery.csv.`

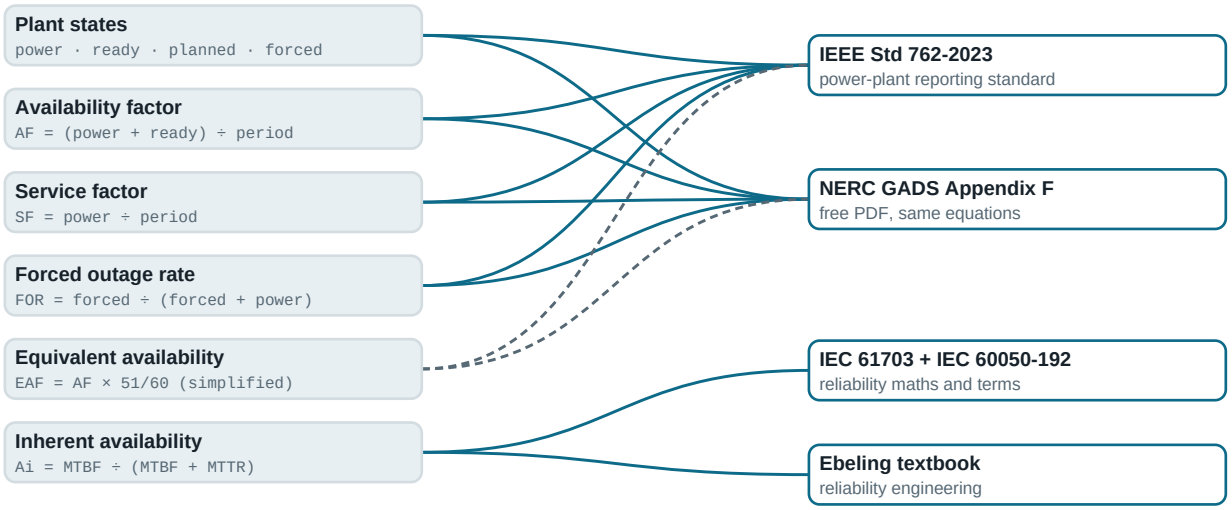
IF YOU REMEMBER ONE THING

Our plant was available about 40 % of logged March hours. The big swing is not availability itself but how the long cool-downs after trips 1, 3 and 8 are classed.

5 · Where the formulas come from

Every number comes from a published power-plant or reliability standard; four choices are our own. The first four numbers follow **IEEE 762**, the standard power plants use to report availability. Inherent availability follows **IEC 61703**, the international standard for reliability maths.

Why it matters: when the result is shown to a customer or partner, they can check each formula against the same source. Where we had to bend a rule to fit an SOFC test plant, it is listed below.



Left: the five numbers (section 3). Right: where each one is defined. Solid line: taken as written. Dashed line: simplified version of the standard formula.

Reference list

#	Reference	What we took from it	Where to get it
1	IEEE Std 762-2023. *IEEE Standard Definitions for Use in Reporting Electric Generating Unit Reliability, Availability, and Productivity.* IEEE, 2023 (replaces IEEE 762-2006).	The four plant states (in service, reserve shutdown, planned outage, forced outage) and the formulas for availability factor, service factor, forced outage rate and equivalent availability.	Paid standard ANSI webstore IEEE 762 working group
2	NERC. *Generating Availability Data System (GADS) Data Reporting Instructions, Appendix F: Performance Indexes and Equations.* North American Electric Reliability Corporation, January 2023.	The same equations as IEEE 762, as used by North American power plants every year. Easiest place to check our formulas.	Free PDF nerc.com
3	IEC 61703:2016 (edition 2). *Mathematical expressions for reliability, availability, maintainability and maintenance support terms.* International Electrotechnical Commission.	Inherent availability from mean time between failures (MTBF) and mean time to repair (MTTR).	Paid standard IEC webstore
4	IEC 60050-192:2015. *International Electrotechnical Vocabulary, Part 192: Dependability.*	Official meaning of the words used in reference 3 (availability, failure, up time, down time).	Free to read online IEC Electropedia
5	C. E. Ebeling. *An Introduction to Reliability and Maintainability Engineering*, 2nd edition. Waveland Press.	Textbook explanation of inherent availability, MTBF and MTTR.	Library or bookshop

6	NREL (National Renewable Energy Laboratory). *Stationary Fuel Cell System Composite Data Products*, and G. Saur et al., *Stationary Fuel Cell Evaluation*, US DOE Annual Merit Review, 2016.	Background only: how availability is reported for fuel cells already in the field. Not used in our calculation.	Free nrel.gov DOE review 2016

References 1 and 3 are paid standards, and their full text was not opened for this guide; the formulas used are the usual published forms. Reference 2 is free and gives the same availability equations, so it is the quickest way to check them.

Where we differ from the standards

1. Shutdown counts as “not available”. In IEEE 762 a unit that is off but could start when asked is in *reserve shutdown* and counts as available. Our plant needs about 13 hours to heat up, so it cannot answer a request quickly; we count Shutdown as planned off. This is stricter than the standard.

2. Hot standby and fuel switch count as “ready”. This matches reserve shutdown in IEEE 762: hot, and able to take load within minutes. (decision C)

3. Equivalent availability uses one fixed factor (51 ÷ 60 kW). The standard adds up the logged hours when the plant could only reach part of its rated power. We have no such log yet, so we scale the whole availability by the estimated best power. (decision D)

4. The strict and lenient rules after a trip are our own. Under the standard's usual reading a forced outage lasts until the unit is available again, which is our strict rule. The lenient rule is an option for when operators decide to cool down anyway. (decision A)

IF YOU REMEMBER ONE THING

Availability factor, service factor, forced outage rate and equivalent availability come from IEEE 762 (free check: NERC GADS Appendix F); inherent availability comes from IEC 61703. Counting Shutdown as not available is our stricter choice.